

The dynamic conditional region: rate, reset work, and the aging context

A problem statement (GO-12)

Geometric Observation program

Problem statement v0.6 — August 2026

Abstract

GO-11 closed the static consumer-relative rate-work region: the $m = 1$ theory in full, the $m = 2$ frontier as a matrix water-level system, and the binary pair in closed form. Every source there was memoryless and every context simultaneous. This problem statement opens the dynamic extension: jointly stationary source-context *processes*, a context that *ages*, and per-symbol rate-work coordinates. The organizing guess is that the water level takes its third promotion — from scalars ($m = 1$), to 2×2 matrices ($m = 2$), to *spectra* (processes) — with the GO-8 staleness law emerging as a limit. Nothing in this document is a claim; the conjectures are gates for future registrations, and the novelty obligations are listed.

1 The problem

Let $(Y_t, V_t)_{t \in \mathbb{Z}}$ be a jointly stationary Gaussian pair process with spectral density matrix $\mathbf{S}(\omega) = \begin{pmatrix} S_Y & S_{YV} \\ S_{YV}^* & S_V \end{pmatrix}$, coherence $\kappa(\omega) = |S_{YV}|^2 / (S_Y S_V)$. The eraser's side information is the *aged, noisy* context

$$S_t = V_{t-\Delta} + U_t, \quad U_t \text{ white, } \text{Var } U = \tau^2, \quad \Delta \geq 0 \text{ the staleness,}$$

observed as the full process S^n . A block code produces the reproduction process $\hat{Y}^n$ with per-symbol Hamming-free (MSE) distortion $\frac{1}{n} \sum E(Y_t - \hat{Y}_t)^2 \leq D$, and the coordinates of interest are the per-symbol limits

$$R(D) = \lim_n \frac{1}{n} I(T^n; \hat{Y}^n), \quad L(D, \Delta) = \lim_n \frac{1}{n} I(T^n; \hat{Y}^n | S^n),$$

with $kT \ln 2 \cdot L$ the ideal per-symbol reset work, exactly as in GO-10/GO-11 but per unit time. Consumers never see S ; Markov $\hat{Y}^n - X^n - S^n$ throughout.

2 Anchors already in the ledger

(i) Δ -degeneracy: for memoryless pairs and $\Delta = 0$ this is GO-11's static theory (Theorems 1–10 there). (ii) The GO-8 staleness law: decode thresholds track an age-dependent discount on AR(1) sources — measured, [replicated] with the 053 caveat — but GO-8 had no exact region behind it; GO-12 is where that region would live. (iii) The work side at $\Delta \rightarrow \infty$ must collapse to the marginal rate-distortion function of the Y process (the context decorrelates), and at $\tau^2 \rightarrow \infty$ likewise.

3 Conjectures

Conjecture 1 (Spectral CR function; the third promotion). *For the rank-one read of the pair process,*

$$L(D, \Delta) = \int_{-\pi}^{\pi} \frac{1}{2} \log_2 g^*(\omega; D(\omega)) \frac{d\omega}{2\pi},$$

where $g^*(\omega; \cdot)$ is GO-11's static quadratic-root CR function applied per frequency with effective parameters $\rho^2(\omega) = \kappa(\omega)$ (staleness entering only through the lag- Δ phase and the noise floor: the conditional spectrum $S_{Y|S}(\omega) = S_Y(1 - \kappa \frac{S_Y}{S_Y + \tau^2})$ is Δ -free for linear aging of this form — the nontrivial content is which aging models break this), and $D(\omega)$ is a distortion allocation determined by two coupled spectral water levels $\gamma_0(\omega), \gamma_1(\omega)$ solving the weighted stationarity system of GO-11's Theorem 3 per frequency under a common pair of Lagrange prices. The $m = 2$ process region takes the same form with the matrix water levels of GO-11's Theorem 9 per frequency.

Theorem 1 (The spectral conditional RDF; work endpoint, circulant embedding; settled and netted, GO-P-2026-070). *Source = the pair process (V, Y) (the formula is provably not the Y -only program); circulant embedding of period n (spectral samples renormalized to mean 1); orthonormal real DFT; per-symbol MSE, i.e. MEAN normalization $\frac{1}{n} \sum_k D_k = D$; records jointly Gaussian (the all-records converse is imported per mode from the static theorem's converse plus the independence/data-processing chain). Then the DFT modes are independent static triples with $S_{Y,k} = \rho^2 \lambda_k + (1 - \rho^2)$, $\rho_k^2 = \rho^2 \lambda_k / S_{Y,k}$, $s_k = 1 + \tau^2 / \lambda_k$ (all machine-verified block-diagonalizations), cross-mode records gain nothing, and*

$$L_n(D) = \min_{\frac{1}{n} \sum_k D_k = D} \frac{1}{n} \sum_k \frac{1}{2} \log_2 g^*(\rho_k^2, s_k, D_k / S_{Y,k}) \xrightarrow{n \rightarrow \infty} \min_{\int D(\omega) \frac{d\omega}{2\pi} = D} \int \frac{1}{2} \log_2 g^*(\rho^2(\omega), s(\omega), D(\omega) / S_Y(\omega)) \frac{d\omega}{2\pi}.$$

The per-mode value φ_k is provably strictly convex decreasing on $(0, S_{Y,k}]$ (verifier proof via the inverted quadratic: $d^2 D / dr^2 = 4x(D' + xD'') > 0$), with a finite left-slope at saturation: mode k saturates iff the water level $\mu < e_k = (s_k - 1) / (2(s_k - 1 + \rho_k^2) S_{Y,k})$ nats; every mode receives $D_k > 0$; the allocation exists always and is unique iff $D < \frac{1}{n} \sum_k S_{Y,k}$. Anchors: $\tau^2 \rightarrow \infty$ recovers classical reverse water-filling of $S_Y(\omega)$ exactly; flat spectrum recovers the static theorem. This is Gray's equal-slope decomposition [16, Thm. 5] pushed to the frequency continuum — the third promotion of the water level, completed for the single-consumer work endpoint. Scoped open: (a) the WEIGHTED version needs two named extra ingredients (Gaussian-scalar optimality and convexity in D_k of the weighted scalar value); (b) the Toeplitz transfer to finite windows is imported-with-lemmas, NOT proven here — (1) joint diagonalization reduces to single-sequence Toeplitz theory (all covariances affine in $T_n(f_V)$, spectra bounded below), (2) continuity of the allocation value under weak spectral convergence, (3) a two-sided sandwich transplanting circulant-optimal records with corner-perturbation control and a blocking lower bound (the conditioning set grows with n), (4) window-vs-infinite-path definition pinning. [R-IND-5: PASS, final report complete — mode mapping to 2.2×10^{-15} ; convexity upgraded from numerical to proven; reduction inequalities re-derived; cross-mode programs at $n = 4, 5$ match the allocation to 10^{-13} including the saturation regime, and the WEIGHTED $w = 0.5$ decomposition confirmed numerically to 1.4×10^{-10} (its two named analytic ingredients remain the open scoping); circulant $\rightarrow$ integral to 2×10^{-16} at $n \geq 64$; classical anchor 4.2×10^{-9} ; finite Toeplitz windows converge $O(1/n)$ from above with n -gap constant to three digits — numerical support for the transfer, not a proof. Harness 070 governed: ALL PASS 4/4, seed 20261001.]

Theorem 2 (The weighted spectral theorem; Conjecture 1 closed at circulant scope; settled and netted, GO-P-2026-071). *Same scope and conventions as Theorem 1, weight $w \in [0, 1]$, records jointly Gaussian (the all-records per-mode import chain is available at every w : the static weighted converse is moment-based and the Markov moment identity holds for any record built from the source plus independent noise). Then cross-mode records gain nothing at every w (chain rule + conditioning inequalities hold for the SAME record, so every convex combination inherits them; $J_{w,k}$ ’s monotonicity closes mean- $\leq$ vs mean- $=$), and*

$$J_{w,n}(D) = \min_{\frac{1}{n} \sum_k D_k = D} \frac{1}{n} \sum_k J_{w,k}(D_k),$$

with $J_{w,k}$ the per-mode static weighted value. $J_{w,k}$ is convex in D_k : the uniqueness Proposition of GO-11 proves the weighted objective convex on the full moment region $\{v > |c|^2\}$ (not merely the active slice — the slice enters only for uniqueness), the moment region is convex with the distortion affine in moments, the moment and channel programs agree at every D (determinant identities + realizability), and the value of a convex program is convex in its right-hand side; the extension past saturation ($\equiv 0$ for $D_k \geq S_{Y,k}$, continuous with finite left-slope) and the affine rescaling $D_k/S_{Y,k}$ preserve convexity; $w = 0$ cites the work-endpoint convexity proof separately (the Proposition covers $w \in (0, 1]$). The allocation is equal-slope with one common distortion price μ ; per mode the optimum carries the two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ of GO-11 Theorem 3 — per-frequency profiles, NOT common quantities. Conjecture 1’s “common pair of Lagrange prices” is thereby refuted as phrased (one distortion constraint, one price) and holds as the pair (scalarization weight w , price μ). Anchors: $w=1$ = classical reverse water-filling of S_Y ; $w=0$ = Theorem 1; $n=1$ = the static weighted value; the value is continuous, increasing, concave in w . Toeplitz transfer: unchanged imported-with-lemmas, with lemma (2) additionally needing continuity of the weighted static value in its parameters (numerically smooth, unique system root everywhere probed). [R-IND-5: PASS — two independent per-mode engines agree to 8.2×10^{-15} over 120 configs; decomposition gaps $\leq 5.6 \times 10^{-16}$ at $n = 4, 5$, two parameter sets, saturated configs closed by direct construction; per-mode two-water-level residuals 2.2×10^{-16} ; single-price reconstruction of the allocation to 2.6×10^{-8} ; anchors to 1.1×10^{-16} .]

Attribution (novelty sweep, 2026-08-05; Gray’s 1972 report page-verified in full): the functional L coincides at the single-letter level with Gray’s conditional rate-distortion function [16, 17], whose operational meaning is side information at both encoder and decoder; existence of the per-symbol limit for stationary-ergodic pairs is Leiner–Gray [18]. The scalar Gaussian evaluation at conditional variance is [16, Thm. 7]; vector-Gaussian conditional/Wyner–Ziv water-filling with a *single* water level over the conditional covariance’s eigenvalues is [19, 20]; single-level spectral water-filling for remote sources is [21]. Our allocation structure generalizes Gray’s equal-slope decomposition [16, Thm. 5] from a discrete conditioning alphabet to the frequency domain. Not found in the sweep: the spectral conditional RDF for jointly stationary Gaussian pairs, the two coupled water levels, the third-party (neither-encoder-nor-consumer) conditioning semantics, or the delayed-noisy side process. Both mandated library pulls are now resolved DOES-NOT-CONTAIN at high confidence (2026-08-05, record in the novelty file): Pinsker’s book contains no fidelity-criterion quantity of any kind (whole-book term-absence tests over the scan’s OCR index; §10.5’s “entropy of one process w.r.t. another” is a KL rate — a terminology trap now on record), and Gray 1973’s own companion report SEL-72-047, read in full, confines the Gaussian content to the scalar pair, cross-referenced as “the examples of [8]”. Residual obligations before final claim wording: an institutional pull of

the T-IT 19(4) text itself (minutes, via the venue package), and the same page-level treatment of Leiner–Gray 1974 [18] plus the Wyner 1976–78 Gaussian-SI line — the natural next places a spectral conditional formula could first appear.

Fact 1 (Δ -invariance under path access; settled and netted, GO-P-2026-065). *In the circulant embedding, with P the cyclic shift and $S^{(\Delta)} = P^\Delta V + U$: the joint laws satisfy the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ (all three cross-covariances match because P commutes with every circulant matrix, e.g. $\text{Cov}(Y, S^{(\Delta)}) = \rho C_V P^{-\Delta}$, and $P^\Delta (C_V + \tau^2 I) P^{-\Delta} = C_V + \tau^2 I$); since P^Δ is a deterministic bijection, $\sigma(S^{(\Delta)})$ carries the same information as $\sigma(S^{(0)})$, so every conditional functional that depends on S only through $\sigma(S)$ — conditional mutual information, the conditional rate–distortion coordinate $L(D)$, conditional entropy, MMSE — is exactly Δ -invariant at every finite n . (Note $\Sigma_{Y|S}$ -invariance alone is weaker: the conditional-mean map does depend on Δ , so label-aligned functionals like $\mathbb{E}[Y_t S_t]$ are not invariant; the recoding identity is the airtight form.) Finite Toeplitz windows deviate only through edge terms, $O(1/n)$ at fixed Δ (empirically $\approx$ linear in Δ over the probed range). Governed harness GO-P-2026-065: circulant invariance to 3.3×10^{-16} , edge ratio 0.5000000000 under n -doubling.*

Attribution (novelty sweeps, 2026-08-05): invariance of conditional functionals under bijective recoding of the conditioning variable is classical folklore, and circulant–Toeplitz $O(1/n)$ asymptotics are standard [11]. The rate-level phenomenon is also in print for memoryless pairs: Simeone and Permuter [1, Lemma 1] note that noncausal availability makes zero-delay conditional RD delay-proof (while positive causal delay makes memoryless side information useless), and Matsuta and Uyematsu [2] characterize when unknown delay leaves the Wyner–Ziv function unchanged. Ours is only the packaged finite- n recoding identity, for a conditioner that is neither encoder nor decoder.

Fact 2 (Slice access: the exact staleness tax, with encoder access width; settled and netted, GO-P-2026-065). *If the eraser instead sees only the time-local context sample $S = V_{t \pm \Delta} + U$ (GO-8’s situation), the single-record conditional coordinate is GO-11’s static quadratic-root CR function with the substitution determined by the encoder’s access: (B) records built from (Y_t, V_t) alone — per-symbol codebooks — pay the quadratic at ρ unchanged, $s \rightarrow s/a^{2\Delta}$ (write $S = a^\Delta V_t + (W + U)$, $W \perp (Y_t, V_t)$, rescale); (A) records granted the context-epoch latent $V_{t \pm \Delta}$ (equivalently, by pair sufficiency, any path encoder with a single-symbol target) pay the strictly smaller value at $\rho \rightarrow \rho a^\Delta$, τ^2 unchanged — the gap reaches 0.053 bits on the probe grid, so the tax has an encoder-side access-width component as well. Both are strictly increasing in Δ ($P(1) < 0$, $P(1/D) > 0$ bracket the root; $dg^*/d\rho^2 < 0$, $dg^*/ds > 0$) with common limit $\frac{1}{2} \log_2(1/D)$. An R-IND-5 pass refuted the first draft of this fact (it asserted the (A) formula for (B) records); the corrected pairing is what the governed harness nets, both substitutions against their own direct channel programs to $\leq 2.4 \times 10^{-13}$.*

Attribution (novelty sweeps, 2026-08-05; mandatory rescope): the substitution device — Markov decay absorbed into the side channel — is published. Simeone and Permuter [1, Eq. (49), test channel Eq. (67)] obtain $R_d(D) = \frac{1}{2} \log_2((1 - \rho^{2d} + \sigma_N^2)/D)$ for a hidden Gauss–Markov source with noiselessly delayed decoder side information — our $s \rightarrow s/a^{2\Delta}$ in a different normalization — and Sun and Cyr [3, Eq. (7)] give the conditional-MI staleness form $-\frac{1}{2} \log_2(1 - a^{2\Delta})$ for AR(1) in the age-of-information literature. In the jointly Gaussian block setting, Wyner–Ziv no-rate-loss additionally makes encoder-side availability of S rate-irrelevant [10]. What is new here is confined to: the noisy delayed slice held by a third-party conditioner, and the encoder-access dichotomy — whether a^Δ enters the read correlation or the effective side-information noise is decided by which variable the record may depend on, which none of the found sources states.

Remark 1. Facts 1–2 settle the causal-origin question the cheap way: the staleness tax is an *access-width* phenomenon — at the eraser (path vs slice) and at the encoder (context-epoch latent vs time- t variables) — not a delay phenomenon. GO-8’s measured age-dependence is the operational face of the slice regime. The two nearest prior statements sit in opposite corners and are both attributed: for *noiseless* Markov sampling the causal path collapses to the freshest sample [3, Lemma 1] (width collapses), and for *memoryless* pairs noncausal access collapses delay [1, Lemma 1]; the dissociation itself — exact zero under path access versus a strictly increasing closed-form tax under slice access, on the same source — is not found in the swept literature.

Remark 2 (Physics scoping). On the thermodynamic side the framing is hemmed by known work, all of which we inherit rather than claim: static conditional erasure work is set by the conditional entropy given a reference (del Rio et al. [12]; Sagawa–Ueda [13]); that retaining stale, non-predictive information is thermodynamically penalized is the “nostalgia” of Still, Sivak, Bell and Crooks [14]; and delay-degraded *work extraction* is quantified in the time-delayed-feedback-demon literature [15]. Our claim is confined to the erasure side of the ledger: the exact dynamic conditional-Landauer curve $W(\Delta)$ for resetting a record against a reference decorrelated by a^Δ , which the sweeps did not find written down. (No co-occurrence of age-of-information and Landauer erasure was found at the arXiv-metadata level as of August 2026; that bridge claim carries a metadata-only caveat.)

Theorem 3 (The conditional-variance reduction; the access family settled in the single-record setting). *Let V be the stationary $AR(1)$ context (pole a , unit variance), $Y_t = \rho V_t + N_t$ (unit variance), observations $S_s = V_s + U_s$, $\text{Var } U = \tau^2 > 0$, and let the record be single-letter (built from (Y_t, V_t) ; the (B) scope of Fact 2), with $D \in (0, 1)$ (the active regime). Then for every context σ -algebra $\mathcal{G}$ generated by a nonempty subset of the observations $\{S_s\}$ — more generally, any $\mathcal{G}$ jointly Gaussian with the V -process and independent of the consumer and record noises, with $q_{\mathcal{G}} < 1$; these hypotheses are essential, since a $\mathcal{G}$ correlated with N_t breaks the sufficiency step —*

$$L_{\mathcal{G}}(D) = \frac{1}{2} \log_2 g^*\left(\rho^2, \frac{1}{1-q_{\mathcal{G}}}, D\right), \quad q_{\mathcal{G}} = \text{Var}(V_t | \mathcal{G}),$$

with $g^(\rho^2, s, D)$ the static quadratic root of GO-11’s Theorem 2: the entire access class enters through one number, the conditional variance of the context at the record epoch. [Proof: $\hat{V} = \mathbb{E}[V_t | \mathcal{G}]$ is a sufficient statistic for $(Y_t, V_t, \hat{\mathbf{Y}})$ within $\mathcal{G}$; rescaling $\hat{V}/(1-q)$ gives an effective side channel $V_t + e'$ with $e' \perp (V_t, N_t)$ jointly Gaussian and $\text{Var } e' = q/(1-q)$, i.e. the static normal form at $s_{\text{eff}} = 1/(1-q)$; the sufficiency step is channel-agnostic — $(T, \hat{\mathbf{Y}}) \perp\!\!\!\perp \mathcal{G} \mid \hat{V}$ holds for every randomized record, so the reduction maps the full problem onto the full static problem and the record-class optimality is imported from the static theorem’s converse, not re-proved here. At $q_{\mathcal{G}} = 1$ (empty access) the continuous extension is $L = \frac{1}{2} \log_2(1/D)$.] Consequently, for the nested access classes at equal lag Δ — slice $\{S_{t+\Delta}\} \subset$ causal prefix $\{S_{\leq t+\Delta}\} \subset$ full path — the coordinates strictly interpolate,*

$$L_{\text{slice}}(D, \Delta) > L_{\text{prefix}}(D, \Delta) > L_{\text{path}}(D) \quad (\tau^2 > 0, a \in (0, 1), \text{ active regime}),$$

with closed/computable forms: $q_{\text{slice}} = 1 - a^{2\Delta}/(1 + \tau^2)$ (recovering Fact 2(B) exactly, $s_{\text{eff}} = (1 + \tau^2)/a^{2\Delta}$); q_{prefix} the scalar steady-state Kalman fixed-lag-smoother variance (Riccati fixed point plus a finite geometric improvement recursion, converging to q_{path} geometrically in Δ); and $q_{\text{path}} = \int S_V \tau^2 / (S_V + \tau^2) \frac{d\omega}{2\pi}$, the noncausal Wiener MMSE. Strictness follows from strict variance reduction under the nesting and $dg^/ds > 0$, and is exact at every finite Δ : the prefix-path gap equals*

$C^{2\Delta}(P_f - q_{\text{path}})$ exactly, with $C = aP_f/P_{\text{pred}} \in (0, 1)$ — never zero, though it falls below double-precision resolution beyond $\Delta \approx \log \varepsilon / \log C^2$. The hypothesis $a \in (0, 1)$ is essential: at $a = 0$ the classes coincide at $\Delta = 0$ and the slice is vacuous for $\Delta \geq 1$. In the thermodynamic reading this is the exact dynamic conditional-Landauer curve: $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}(D, \Delta)$ falls geometrically from the filtering-set cost to the smoothing-set floor as the eraser is allowed to wait. [Pre-assertion numerics, seed 20260916: reduction vs direct channel optimization to 1.4×10^{-13} on four access classes; Riccati and Wiener forms vs windowed covariances to 2×10^{-16} ; interpolation strict at lags 0, 1, 3, 6 and at a second instance.]

Conjecture 2 (The process-rate causal object; the open residue). *Theorem 3 settles the causal-prefix eraser in the single-record setting, where the Kalman filtering-vs-smoothing gap carries the interpolation — as the sweep’s caveat anticipated. What remains open is the process-rate version: block records amortized across symbols, with the per-symbol coordinate defined via Markov-constrained causal conditioning in the sense of Lev–Khina [5] — a deliberate choice, since the Kostina–Hassibi causally-conditioned-DI definition [6] provably collapses to the full-path endpoint in Gaussian settings. The nearest known object (causal decoder side information, undelayed) is explicitly open even there [5]; the vector NRDF water-filling caution [9] applies; and the noiseless endpoint collapses by [3, Lemma 1], so any interpolation theorem must be carried by $\tau^2 > 0$.*

Conjecture 3 (Spectral tilt). *The encoder-side tilt (GO-11, 061/062) becomes spectral: the L -optimal record concentrates its precision in frequency bands where the context coherence $\kappa(\omega) \frac{S_V}{S_V + \tau^2}$ is largest — records written in the bands the eraser’s context can read — with an analytically-zero control in any band of zero coherence.*

4 Questions

Question 1. *Does Gray’s conditional-RD theory for stationary sources (or any successor) already contain Conjecture 1’s noncausal form? Novelty sweep obligation: conditional/two-sided RD for processes, remote-source RD with delayed side information, causal RD (Neuhoff–Gilbert lineage), zero-delay work. No claim before the sweep.*

Question 2. *Is there a process analog of Theorem 10 (binary pair process with Markov flips) in closed form, or does the discrete side stop at numerics?*

Question 3. *What is the right operational face? Candidate: 058-lineage decode thresholds on materialized codebooks for AR(1) pairs at three staleness values, with the noncausal-zero control of Fact 1 as the physics gate separating path from slice erasure, and the (A)-vs-(B) encoder gap of Fact 2 as a second measurable face.*

5 Verification obligations

House rules apply unchanged: R-IND-5 fresh-context verification before any assertion; novelty sweeps before any novelty language; PROTOCOL §5.1 power/pilot fields at any seal; every eventual theorem harness-netted and CI-enforced. The first registration (GO-P-2026-065, sealed and run at v0.2) settled the opening control: Facts 1–2 passed their governed harness ALL PASS 4/4 (seed 20260915), after an R-IND-5 pass refuted and corrected the slice fact’s first draft. The open theory

objects are Conjectures 1 and 2. All four novelty sweeps are complete (~ 63 queries, 10 page-verified documents including Gray’s 1972 report in full; record in `go12-process-region-NOVELTY.md`), with attributions folded above at v0.3. Outstanding sweep debts, on record: page-level reads of [17] and of Pinsker’s conditional ε -entropy chapter before Conjecture 1 carries novelty language, and the stochastic-thermodynamics information-flow flank (Hartich–Barato–Seifert; Horowitz–Esposito) before the *causal*-measurement physics framing does.

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